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AegisLife Insurance Analytics & Risk Assessment

Tools Used:

- Python (Pandas, NumPy, Seaborn, Matplotlib)
- MySQL
- Power BI
- Excel

Business Problem & Objectives

Business Problem

AegisLife Insurance wants to analyze customer behavior, claims, fraud risk, policy performance, and agent productivity to improve profitability and operational efficiency.

Objectives

- Analyze customer demographics and risk profiles
- Evaluate claim performance and approval rates
- Identify fraud patterns
- Assess policy and product profitability
- Measure agent productivity
- Build interactive dashboards for decision-making

Data Preparation & SQL Integration

Datasets Used

Table	Records
Customer Master	1,648
Policy Details	2,828
Claim History	1,406
Agent Info	300
Customer Feedback	1,000

Data Cleaning Performed

- Removed status/helper columns
- Standardized column names
- Corrected category inconsistencies
 - Properties → Property
 - Mal → Male
 - males → Male
 - females → Female
 - souht → South
 - Wset → West
 - Aproval → Approved
 - Rejeted → Rejected
- Validated primary keys
- Validated foreign key relationships
- Checked duplicates and missing values

SQL Validation

- No duplicate IDs found
- No invalid customer references
- No invalid policy references
- No invalid agent references
- No invalid claim references

Key Business Insights

Portfolio Overview

- Total Customers: 1,648
- Total Policies: 2,828
- Total Claims: 1,406
- Total Claim Amount: ₹355.62M

Claims & Risk

- Claim Approval Rate: 35.85%
- Fraud Rate: 49%
- Loss Ratio: 4.71
- Policy Lapse Rate: 32.32%

Product Analysis

- Vehicle Insurance has highest policy count (591)
- Health Insurance generates highest claim amount
- Property Insurance has highest loss ratio

Regional Analysis

- Central region has highest total claim amount
- South region has highest loss ratio

Agent Performance

- Average policies sold per agent: 176
- Average fraud association: 4.56
- Average premium sold: ₹27.66K

Hypothesis Testing

Test 1: Smokers vs Non-Smokers Claim Amount

Result:

- p-value = 0.0598

Conclusion:

No statistically significant difference in claim amount between smokers and non-smokers.

Test 2: Claim Amount by Region

Result:

- p-value = 0.5762

Conclusion:

Average claim amounts do not significantly differ across regions.

Test 3: Claim Type vs Claim Status

Result:

- p-value = 0.6156

Conclusion:

Claim status is independent of claim type.

Test 4: Risk Score vs Claim Amount

Result:

- Correlation = -0.0157
- p-value = 0.5572

Conclusion:

No significant relationship exists between risk score and claim amount.

Recommendations & Conclusion

Recommendations

1. Strengthen fraud detection processes due to high fraud rate (49%).
2. Review Property Insurance pricing and underwriting policies.
3. Improve customer retention to reduce policy lapse rate.
4. Investigate reasons behind low claim approval rates.
5. Monitor high-risk customers more closely.
6. Benchmark agent performance using portfolio averages.

Conclusion

The analysis identified key drivers of insurance performance across customers, products, claims, and agents. While policy sales are balanced across products, high loss ratios, fraud exposure, and policy lapses present major business challenges. The Power BI dashboards provide actionable insights to support data-driven decision-making and risk management.